TEJAS MANDWADE

Python Developer | Backend Developer | AI Engineer

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Professional Summary

Python focused Software Engineer passionate about building production-grade backend and AI applications using Python, FastAPI, SQL, and modern AI frameworks. Designed and developed VitalFlow, an AI-powered clinical intelligence platform predicting 41 medical conditions with Explainable AI and multimodal voice/image support, and HireMind, an offline recruitment intelligence platform processing 100,000+ candidate profiles through semantic matching and multi-signal ranking. Skilled in REST APIs, Machine Learning, NLP, Explainable AI (XAI), model calibration, scalable backend architecture, and cloud deployment.

Education

Godavari College of Engineering, Jalgaon | DBATU — **Expected Jul 2026**

B.Tech in Computer Engineering

HSC — Sane Guruji Sec. & Higher Sec. School, Yawal — **78%**

SSC — K.Narkhede Vidyalaya & Kanishtha Mahavidyalaya, Bhusawal — **75.20%**

Experience

Edunet Foundation — Feb 2025 – Mar 2025

Artificial Intelligence Intern

- Conceptualized an AI healthcare diagnostic system during the internship later independently architected and extended into VitalFlow, a full scale multimodal clinical intelligence platform.
- Developed ML prototypes using Scikit-learn and built Python based AI pipelines (data validation, model training, API integration) forming the technical foundation for VitalFlow's diagnostic engine.

Projects

VitalFlow — AI-Powered Clinical Intelligence Platform

Python · Scikit-learn · Groq API · Gradio · SQLite

- Architected an end-to-end clinical decision support application predicting 41 medical conditions from 132 structured symptom features using a Calibrated Random Forest model.
- Built an Explainable AI (XAI) module highlighting the most influential symptoms behind each prediction and generating differential diagnosis comparisons for improved model transparency.
- Designed a rule based Clinical Risk Engine detecting critical symptoms (e.g., chest pain, severe bleeding), bypassing ML inference to prioritize emergency recommendations.
- Integrated Groq Whisper (speech-to-text) and Llama 3 Vision (image understanding) for multimodal symptom collection, and built automated PDF reports (fpdf2) with SQLite logged prediction history.

HireMind — Offline AI Recruitment & Candidate Ranking Platform

Python · Streamlit · Sentence Transformers · Pandas · NumPy · Pytest

- Built an offline AI recruitment platform processing and ranking 100,000+ candidate profiles via semantic skill matching and multi-signal evaluation, with no dependency on external LLM APIs.
- Designed a 5-signal ranking engine (Role Fit, Semantic Skills, Hiring Intent, Career Growth, Stability) using Sentence Transformers (all-MiniLM-L6-v2) to generate a recruiter-ready Top 100 shortlist with explainable scoring.
- Engineered a data validation pipeline that filtered 18,899 suspicious profiles and built an interactive Streamlit dashboard with score breakdowns, recruiter verdicts, and Pytest backed testing.

Full-Stack Agentic AI Platform

Python · FastAPI · LangGraph · LangChain · Streamlit

- Built a full-stack Agentic AI platform integrating Groq, OpenAI, and Tavily for real time AI responses.
- Designed a ReAct-based workflow via FastAPI REST APIs for tool augmented, multichat, voice conversations.

Technical Skills

Languages: Python, SQL, JavaScript
Backend: FastAPI, RESTful APIs, API Integration
AI & Machine Learning: Scikit-learn, LangChain, LangGraph, Sentence Transformers, Prompt Engineering, NLP, XAI, Model Calibration
Databases: PostgreSQL, SQLite
Tools: Git, GitHub, Streamlit, Gradio, Pandas, NumPy, Pytest
Cloud & AI Services: Render, Hugging Face Spaces, Groq API, OpenAI API, Tavily API